Grade 11 – Term 1
C9: Normal versus Student's t, with Errors

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1 Models and calculation tools

Open the normal and Student's t reference tables.

Page 1 gives $\Phi(z) = P(Z \leq z)$ for positive z on a hundredth grid (through 2.59). Page 2 gives positive t critical values for upper-tail probabilities $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$, for df 1–30 and selected larger df. Those entries do not give arbitrary CDF values. Use a suitable calculator's t CDF and inverse t CDF for this activity; normal CDF and inverse normal are also allowed. Numerical CDF and quantile results in the solutions are supplementary to the tables. Never turn table bounds into a single probability or error.

Write $F_\nu(t) = P(T \leq t)$ for $T \sim t_\nu$. Symmetry gives $F_\nu(-c) = 1 - F_\nu(c)$; a right tail is $1 - F_\nu(c)$. An interval has probability $F_\nu(b) - F_\nu(a)$ and its outside complement has probability $F_\nu(a) + 1 - F_\nu(b)$. Replace F_ν by Φ for normal calculations. Endpoints have zero probability. For central area p, each tail has $(1-p)/2$; for combined symmetric tails q, each tail has $q/2$.

A contextual location-scale model is explicitly $X = m + aT$, with $a > 0$ a scale parameter; a is not automatically the standard deviation. Thus $T = (X - m)/a$. These are stipulated models for signed deviations, not claims that arbitrary individual measurements follow t. For independent random observations from a normal population, with unknown population standard deviation, the sampling statistic instead is

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Here $\bar{X}$ and S vary across samples; do not treat observed s as a fixed population standard deviation.

Attempt each task before checking the worked solution. Calculate with full precision, then report probabilities to four decimals, standardized quantiles to three decimals, contextual values to two decimals, and percentage errors to two decimals. Displayed rounded intermediate values are not the inputs used for errors. Diagrams shade the requested region; curves continue beyond the displayed range.

C9. Compare normal approximations with t probabilities and quantiles, and interpret their errors. For each comparison define $Z \sim N(0,1)$ and $T \sim t_\nu$. The normal calculation approximates the t calculation. In contextual units use $X_N = m + aZ$ and $X_T = m + aT$, with exactly the same m and a. Probability comparisons keep the same boundaries and event; inverse comparisons keep the same target probability. With normal result r_N and nonzero t reference r_T, calculate

$$A = |r_N - r_T|, \qquad R = \frac{|r_N - r_T|}{|r_T|}, \qquad E = 100R\%.$$

For intervals compare each endpoint separately. These errors depend on the quantity compared: a standardized quantile and a contextual cutoff need not have the same relative error. Heavier t tails give larger probabilities beyond positive cutoffs and more extreme quantiles at the same tail probability. As df increases, t approaches normal. An interval entirely on one side of zero need not contain less t probability.

2 Direct probability problems

Problems:

2.1. Delivery schedule deviations

2.2. Production cost deviations

2.3. Electricity forecast deviations

2.4. Monthly sales deviations

2.5. Commute forecast deviations

2.6. Revenue planning alerts

2.1 Delivery schedule deviations

Problem. A planning team records how far delivery time is above its scheduled time, in minutes; a result below its scheduled time is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed nine on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by five to give the difference in minutes.

Questions/Tasks

C9 Find the chance of a delivery arriving less than seven and a half minutes late, including early arrivals. Also find the chance of arriving more than six minutes early. Use both prediction rules for each chance. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

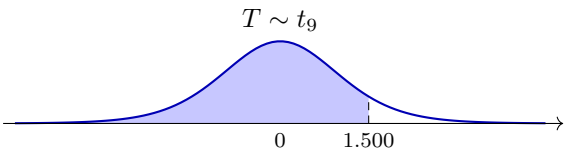
Let X_T and X_N denote delivery time minus its scheduled time, in minutes. The stated table patterns specify $T \sim t_9$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 5T$, $X_N = 0 + 5Z$. Thus $m = 0$ and $a = 5$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/5$. The requested events are $P(X < 7.5)$, $P(X < -6)$, with $X = X_T$ or X_N for the appropriate model. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

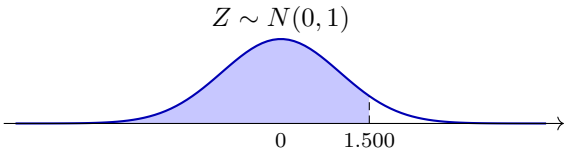
Mathematical reasoning and calculations

Standardize each boundary using $(x-0)/5$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(X_T < 7.5) = P(T < 1.5).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). The region is cumulative area to the left. Supplementary CDF values: $F_9(1.5) \approx 0.9160747$.

$$P(T < 1.5) = F_9(1.5) \approx 0.9161.$$

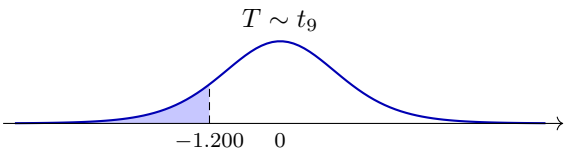
$$P(Z < 1.5) = \Phi(1.5) \approx 0.9332.$$

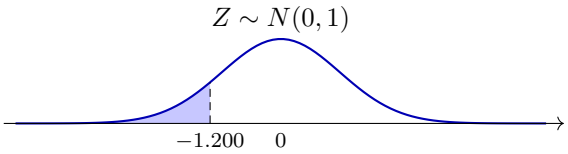
$$A = |0.9331928 - (0.9160747)| \approx 0.017118, \quad R = \frac{A}{0.9160747} \approx 0.018686, \quad E \approx 1.87\%.$$

Interpretation. About 91.61% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 1.71 percentage points (1.87% relative to t).

Part 2.

$$P(X_T < -6) = P(T < -1.2).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). The region is cumulative area to the left. By symmetry, $F_9(-1.2) = 1 - F_9(1.2) \approx 1 - 0.8696134 = 0.1303866$. Supplementary CDF values: $F_9(-1.2) \approx 0.1303866$.

$$P(T < -1.2) = F_9(-1.2) \approx 0.1304.$$

$$P(Z < -1.2) = \Phi(-1.2) \approx 0.1151.$$

$$A = |0.1150697 - (0.1303866)| \approx 0.015317, \quad R = \frac{A}{0.1303866} \approx 0.117473, \quad E \approx 11.75\%.$$

Interpretation. About 13.04% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 1.53 percentage points (11.75% relative to t).

2.2 Production cost deviations

Problem. A planning team records how far batch cost is above its budget, in monetary units; a result below its budget is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed fourteen on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by twenty to give the difference in monetary units.

Questions/Tasks



C9 Find the chance of a batch costing more than thirty monetary units above budget. Also find the chance of a batch costing more than the amount that is sixteen monetary units below budget. Use both prediction rules for each chance. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

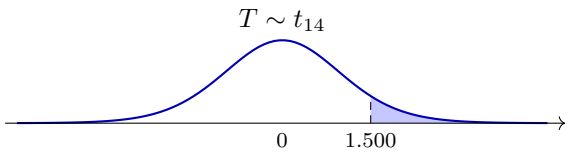
Let X_T and X_N denote batch cost minus its budget, in monetary units. The stated table patterns specify $T \sim t_{14}$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 20T$, $X_N = 0 + 20Z$. Thus $m = 0$ and $a = 20$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/20$. The requested events are $P(X > 30)$, $P(X > -16)$, with $X = X_T$ or X_N for the appropriate model. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

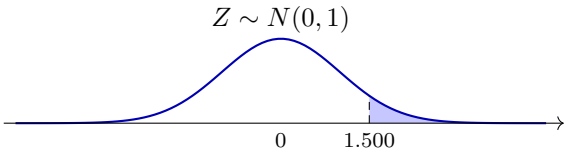
Mathematical reasoning and calculations

Standardize each boundary using $(x - 0)/20$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(X_T > 30) = P(T > 1.5).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Take the complement of the cumulative area. Supplementary CDF values: $F_{14}(1.5) \approx 0.9220873$.

$$P(T > 1.5) = 1 - F_{14}(1.5) \approx 0.0779.$$

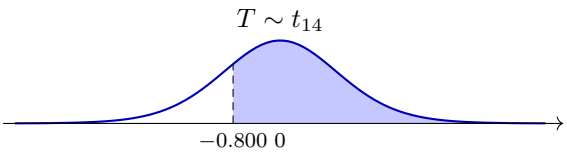
$$P(Z > 1.5) = 1 - \Phi(1.5) \approx 0.0668.$$

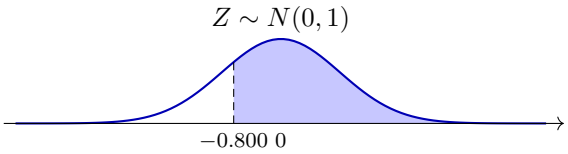
$$A = |0.0668072 - (0.0779127)| \approx 0.011105, \quad R = \frac{A}{0.0779127} \approx 0.142537, \quad E \approx 14.25\%.$$

Interpretation. About 7.79% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 1.11 percentage points (14.25% relative to t).

Part 2.

$$P(X_T > -16) = P(T > -0.8).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Take the complement of the cumulative area. By symmetry, $F_{14}(-0.8) = 1 - F_{14}(0.8) \approx 1 - 0.7814622 = 0.2185378$. Supplementary CDF values: $F_{14}(-0.8) \approx 0.2185378$.

$$P(T > -0.8) = 1 - F_{14}(-0.8) \approx 0.7815.$$

$$P(Z > -0.8) = 1 - \Phi(-0.8) \approx 0.7881.$$

$$A = |0.7881446 - (0.7814622)| \approx 0.006682, \quad R = \frac{A}{0.7814622} \approx 0.008551, \quad E \approx 0.86\%.$$

Interpretation. About 78.15% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 0.67 percentage points (0.86% relative to t).

2.3 Electricity forecast deviations

Problem. A planning team records how far household electricity use is above its forecast, in kilowatt-hours; a result below its forecast is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed nineteen on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by fifty to give the difference in kilowatt-hours.

Questions/Tasks

C9 Find the chance that household use is more than twelve and a half but less than fifty-seven and a half kilowatt-hours above forecast. Use both prediction rules for each chance. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

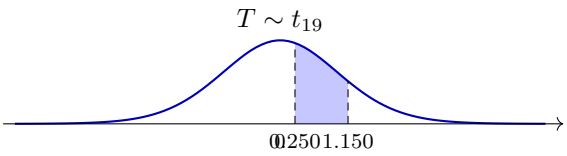
Let X_T and X_N denote household electricity use minus its forecast, in kilowatt-hours. The stated table patterns specify $T \sim t_{19}$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 50T$, $X_N = 0 + 50Z$. Thus $m = 0$ and $a = 50$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/50$. The requested events are $P(12.5 < X < 57.5)$, with $X = X_T$ or X_N for the appropriate model. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

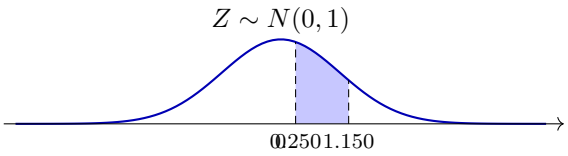
Mathematical reasoning and calculations

Standardize each boundary using $(x - 0)/50$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(12.5 < X_T < 57.5) = P(0.25 < T < 1.15).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Subtract the left cumulative area from the right cumulative area. Supplementary CDF values: $F_{19}(0.25) \approx 0.5973648$, $F_{19}(1.15) \approx 0.8677933$.

$$P(0.25 < T < 1.15) = F_{19}(1.15) - F_{19}(0.25) \approx 0.2704.$$

$$P(0.25 < Z < 1.15) = \Phi(1.15) - \Phi(0.25) \approx 0.2762.$$

$$A = |0.2762217 - (0.2704285)| \approx 0.005793, \quad R = \frac{A}{0.2704285} \approx 0.021422, \quad E \approx 2.14\%.$$

Interpretation. About 27.04% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 0.58 percentage points (2.14% relative to t).

2.4 Monthly sales deviations

Problem. A planning team records how far monthly sales is above its target, in units sold; a result below its target is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed eleven on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by twenty to give the difference in units sold.

Questions/Tasks

C9 Find the chance that monthly sales are more than seventeen but less than thirty-six units below target. Use both prediction rules for each chance. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

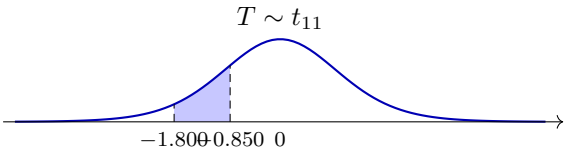
Let X_T and X_N denote monthly sales minus its target, in units sold. The stated table patterns specify $T \sim t_{11}$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 20T$, $X_N = 0 + 20Z$. Thus $m = 0$ and $a = 20$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/20$. The requested events are $P(-36 < X < -17)$, with $X = X_T$ or X_N for the appropriate model. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

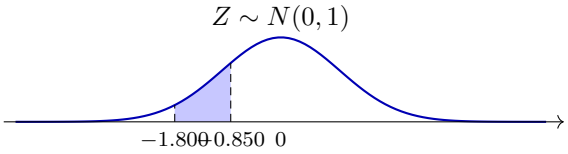
Mathematical reasoning and calculations

Standardize each boundary using $(x - 0)/20$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(-36 < X_T < -17) = P(-1.8 < T < -0.85).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{11}(-1.8) = 1 - F_{11}(1.8) \approx 1 - 0.9503421 = 0.0496579$. By symmetry, $F_{11}(-0.85) = 1 - F_{11}(0.85) \approx 1 - 0.7932783 = 0.2067217$. Supplementary CDF values: $F_{11}(-1.8) \approx 0.0496579$, $F_{11}(-0.85) \approx 0.2067217$.

$$P(-1.8 < T < -0.85) = F_{11}(-0.85) - F_{11}(-1.8) \approx 0.1571.$$

$$P(-1.8 < Z < -0.85) = \Phi(-0.85) - \Phi(-1.8) \approx 0.1617.$$

$$A = |0.1617322 - (0.1570638)| \approx 0.004668, \quad R = \frac{A}{0.1570638} \approx 0.029723, \quad E \approx 2.97\%.$$

Interpretation. About 15.71% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 0.47 percentage points (2.97% relative to t).

2.5 Commute forecast deviations

Problem. A planning team records how far commute time is above its forecast, in minutes; a result below its forecast is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed twenty-four on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by ten to give the difference in minutes.

Questions/Tasks

C9 Find the chance that a commute finishes after the point nine and a half minutes earlier than forecast and before the point ten and a half minutes later than forecast. Use both prediction rules for each chance. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

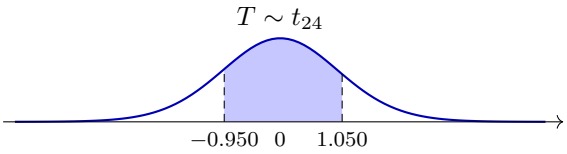
Let X_T and X_N denote commute time minus its forecast, in minutes. The stated table patterns specify $T \sim t_{24}$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 10T$, $X_N = 0 + 10Z$. Thus $m = 0$ and $a = 10$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/10$. The requested events are $P(-9.5 < X < 10.5)$, with $X = X_T$ or X_N for the appropriate model. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

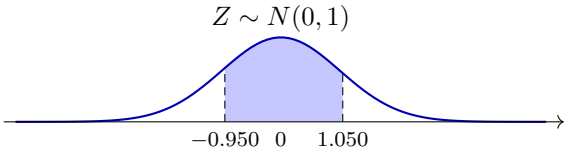
Mathematical reasoning and calculations

Standardize each boundary using $(x - 0)/10$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(-9.5 < X_T < 10.5) = P(-0.95 < T < 1.05).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{24}(-0.95) = 1 - F_{24}(0.95) \approx 1 - 0.8242092 = 0.1757908$. Supplementary CDF values: $F_{24}(-0.95) \approx 0.1757908$, $\quad F_{24}(1.05) \approx 0.8479116$.

$$P(-0.95 < T < 1.05) = F_{24}(1.05) - F_{24}(-0.95) \approx 0.6721.$$

$$P(-0.95 < Z < 1.05) = \Phi(1.05) - \Phi(-0.95) \approx 0.6821.$$

$$A = |0.6820848 - (0.6721208)| \approx 0.009964, \quad R = \frac{A}{0.6721208} \approx 0.014825, \quad E \approx 1.48\%.$$

Interpretation. About 67.21% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 1.00 percentage points (1.48% relative to t).

2.6 Revenue planning alerts

Problem. A planning team records how far weekly revenue is above its plan, in thousands of monetary units; a result below its plan is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed five on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by eight to give the difference in thousands of monetary units.

Questions/Tasks



C9 Find the chance that weekly revenue is more than eight thousand four hundred monetary units below plan or more than ten thousand eight hundred above plan. Also find the chance that revenue is more than sixteen thousand monetary units away from plan in either direction. Use both prediction rules for each chance. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

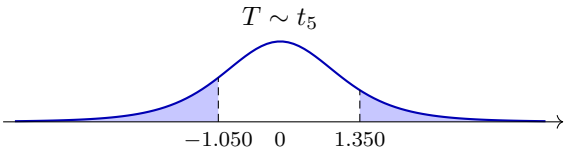
Let X_T and X_N denote weekly revenue minus its plan, in thousands of monetary units. The stated table patterns specify $T \sim t_5$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 8T$, $X_N = 0 + 8Z$. Thus $m = 0$ and $a = 8$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/8$. The requested events are $P(X < -8.4 \text{ or } X > 10.8)$, $P(X < -16 \text{ or } X > 16)$, with $X = X_T$ or X_N for the appropriate model. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

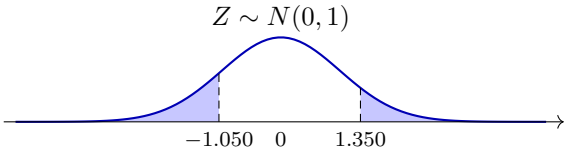
Mathematical reasoning and calculations

Standardize each boundary using $(x-0)/8$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(X_T < -8.4 \text{ or } X_T > 10.8) = P(T < -1.05 \text{ or } T > 1.35).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Add the disjoint outside regions, or subtract the intervening interval from one. By symmetry, $F_5(-1.05) = 1 - F_5(1.05) \approx 1 - 0.8291023 = 0.1708977$. Supplementary CDF values: $F_5(-1.05) \approx 0.1708977$, $\quad F_5(1.35) \approx 0.8825456$.

$$P(T < -1.05 \text{ or } T > 1.35) = F_5(-1.05) + 1 - F_5(1.35) \approx 0.2884.$$

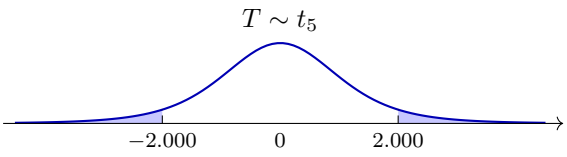
$$P(Z < -1.05 \text{ or } Z > 1.35) = \Phi(-1.05) + 1 - \Phi(1.35) \approx 0.2354.$$

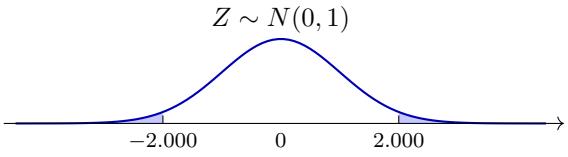
$$A = |0.2353670 - (0.2883521)| \approx 0.052985, \quad R = \frac{A}{0.2883521} \approx 0.183751, \quad E \approx 18.38\%.$$

Interpretation. About 28.84% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 5.30 percentage points (18.38% relative to t).

Part 2.

$$P(X_T < -16 \text{ or } X_T > 16) = P(T < -2 \text{ or } T > 2).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Add the disjoint outside regions, or subtract the intervening interval from one. By symmetry, $F_5(-2) = 1 - F_5(2) \approx 1 - 0.9490303 = 0.0509697$. Supplementary CDF values: $F_5(-2) \approx 0.0509697, \quad F_5(2) \approx 0.9490303$.

$$P(T < -2 \text{ or } T > 2) = F_5(-2) + 1 - F_5(2) \approx 0.1019.$$

$$P(Z < -2 \text{ or } Z > 2) = \Phi(-2) + 1 - \Phi(2) \approx 0.0455.$$

$$A = |0.0455003 - (0.1019395)| \approx 0.056439, \quad R = \frac{A}{0.1019395} \approx 0.553654, \quad E \approx 55.37\%.$$

Interpretation. About 10.19% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 5.64 percentage points (55.37% relative to t).

3 Inverse probability problems

Problems:

3.1. Top delivery delays

3.2. Bottom cost deviations

3.3. Central energy deviations

3.4. Central sales deviations

3.5. Two-tail checks on filling weights

3.6. Two-tail checks on processing times

3.1 Top delivery delays

Problem. A planning team records how far delivery time is above its scheduled time, in minutes; a result below its scheduled time is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed nine on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by five to give the difference in minutes. Find the lateness exceeded by only five percent of deliveries.

Questions/Tasks



C9 Describe the outcomes being counted and the share at or below each limit. Find the requested limits using both rules, first as readings before multiplication and then in the original units. Compare each pair of corresponding readings before multiplication separately. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

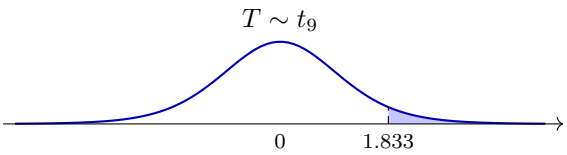
Let X_T and X_N denote delivery time minus its scheduled time, in minutes. The stated table patterns specify $T \sim t_9$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0{+}5T$, $X_N = 0{+}5Z$. Thus $m = 0$ and $a = 5$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x-0)/5$. The target is $P(X_T > x_T) = P(X_N > x_N) = 0.05$. Here x_T, x_N are contextual cutoffs and c_T, c_N are standardized cutoffs. For symmetric intervals the contextual endpoints are $-ac$ and ac under each model. Compare corresponding standardized endpoints separately. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

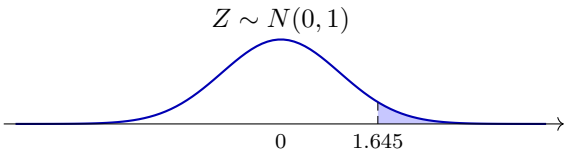
Mathematical reasoning and calculations

$$P(X_T > x) = 0.0500 \quad \Longrightarrow \quad P(T \le c) = 0.9500, \qquad c = \frac{x-0}{5}.$$

A top percentage is a right tail, so subtract it from one before inverse CDF.

$$c_T = F_9^{-1}(0.95) \approx 1.833, \qquad c_N = \Phi^{-1}(0.95) \approx 1.645.$$





$$A = |1.6448536 - (1.8331129)| \approx 0.188259, \quad R = \frac{A}{1.8331129} \approx 0.102699, \quad E \approx 10.27\%.$$

$$x_T = 0 + 5c_T \approx 9.17 \text{ minutes}, \qquad x_N = 0 + 5c_N \approx 8.22 \text{ minutes}.$$

Interpretation. About 5% of deviations are above 9.17 minutes. Normal places the cutoff closer to the location, so it understates the magnitude of this extreme deviation. Errors above compare standardized quantiles.



3.2 Bottom cost deviations

Problem. A planning team records how far batch cost is above its budget, in monetary units; a result below its budget is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed eleven on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by twenty to give the difference in monetary units. Find the cost difference below which ten percent of batches fall.

Questions/Tasks

C9 Describe the outcomes being counted and the share at or below each limit. Find the requested limits using both rules, first as readings before multiplication and then in the original units. Compare each pair of corresponding readings before multiplication separately. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

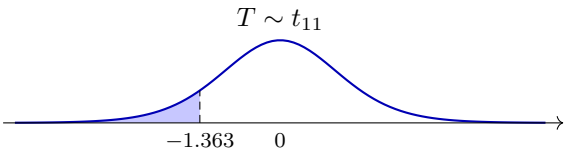
Let X_T and X_N denote batch cost minus its budget, in monetary units. The stated table patterns specify $T \sim t_{11}$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 20T$, $X_N = 0 + 20Z$. Thus $m = 0$ and $a = 20$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/20$. The target is $P(X_T < x_T) = P(X_N < x_N) = 0.10$. Here x_T, x_N are contextual cutoffs and c_T, c_N are standardized cutoffs. For symmetric intervals the contextual endpoints are $-ac$ and ac under each model. Compare corresponding standardized endpoints separately. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

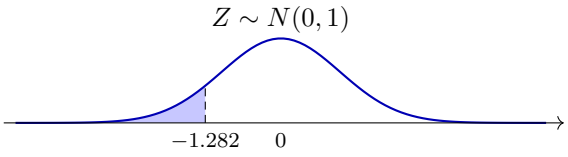
Mathematical reasoning and calculations

$$P(X_T < x) = 0.1000 \quad \Longrightarrow \quad P(T \le c) = 0.1000, \qquad c = \frac{x - 0}{20}.$$

The bottom percentage is already a left cumulative probability; the cutoff is negative by symmetry.

$$c_T = F_{11}^{-1}(0.1) \approx -1.363, \qquad c_N = \Phi^{-1}(0.1) \approx -1.282.$$





$$A = |-1.2815516 - (-1.3634303)| \approx 0.081879, \quad R = \frac{A}{1.3634303} \approx 0.060053, \quad E \approx 6.01\%.$$

$$x_T = 0 + 20c_T \approx -27.27 \text{ monetary units}, \qquad x_N = 0 + 20c_N \approx -25.63 \text{ monetary units}.$$

Interpretation. About 10% of deviations are below -27.27 monetary units. Normal places the cutoff closer to the location, so it understates the magnitude of this extreme deviation. Errors above compare standardized quantiles

3.3 Central energy deviations

Problem. A planning team records how far household electricity use is above its forecast, in kilowatt-hours; a result below its forecast is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed nineteen on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by fifty to give the difference in kilowatt-hours. Find two limits equally far below and above the forecast that enclose the middle ninety-five percent of differences in electricity use.

Questions/Tasks

C9 Describe the outcomes being counted and the share at or below each limit. Find the requested limits using both rules, first as readings before multiplication and then in the original units. Compare each pair of corresponding readings before multiplication separately. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

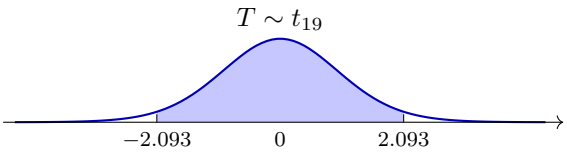
Let X_T and X_N denote household electricity use minus its forecast, in kilowatt-hours. The stated table patterns specify $T \sim t_{19}$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 50T$, $X_N = 0 + 50Z$. Thus $m = 0$ and $a = 50$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/50$. The target is $P(-c_T < T < c_T) = P(-c_N < Z < c_N) = 0.95$. Here x_T, x_N are contextual cutoffs and c_T, c_N are standardized cutoffs. For symmetric intervals the contextual endpoints are $-ac$ and ac under each model. Compare corresponding standardized endpoints separately. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

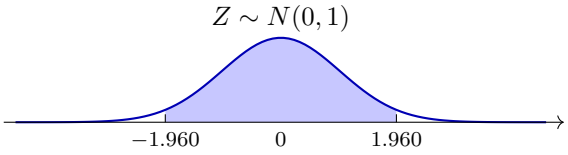
Mathematical reasoning and calculations

$$P(-c < T < c) = 0.9500 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0250.$$

Divide the outside complement $1 - p$ by two because symmetry makes the tails equal.

$$F_{19}(c) = 0.9750, \qquad c_T = F_{19}^{-1}(0.975) \approx 2.093, \quad c_N = \Phi^{-1}(0.975) \approx 1.960.$$





The standardized endpoints are $[-2.093, 2.093]$ under t and $[-1.960, 1.960]$ under normal. Compare the corresponding lower endpoints:

$$A = |-1.9599640 - (-2.0930241)| \approx 0.133060, \quad R = \frac{A}{2.0930241} \approx 0.063573, \quad E \approx 6.36\%.$$

Compare the corresponding upper endpoints:

$$A = |1.9599640 - (2.0930241)| \approx 0.133060, \quad R = \frac{A}{2.0930241} \approx 0.063573, \quad E \approx 6.36\%.$$

$$L_T = 0 - 50c_T \approx -104.65, \qquad U_T = 0 + 50c_T \approx 104.65 \quad \text{(kWh)}.$$

Normal gives $[-98.00, 98.00]$ kWh. Errors above compare standardized endpoints, not these contextual values. **Interpretation.** The t interval contains 95% of modeled deviations; 2.5% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.



3.4 Central sales deviations

Problem. A planning team records how far monthly sales is above its target, in units sold; a result below its target is recorded as below zero. The team agrees to use two prediction rules for these differences. The first uses the full pattern of possible readings behind the row headed twenty-nine on the second page of the supplied reference tables. The second uses the full bell-shaped pattern behind the first page. For both rules, readings are centered on zero, and each reading is multiplied by twenty to give the difference in units sold. Find two limits equally far below and above the sales target that enclose the middle eighty percent of sales differences.

Questions/Tasks



C9 Describe the outcomes being counted and the share at or below each limit. Find the requested limits using both rules, first as readings before multiplication and then in the original units. Compare each pair of corresponding readings before multiplication separately. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

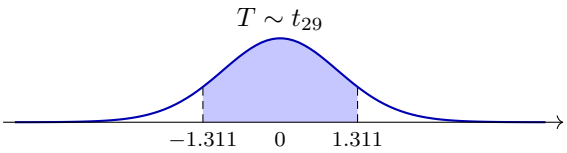
Let X_T and X_N denote monthly sales minus its target, in units sold. The stated table patterns specify $T \sim t_{29}$ and $Z \sim N(0,1)$, and the conversion rules specify $X_T = 0 + 20T$, $X_N = 0 + 20Z$. Thus $m = 0$ and $a = 20$; the scale is not a stipulated standard deviation. A contextual boundary x becomes $c = (x - 0)/20$. The target is $P(-c_T < T < c_T) = P(-c_N < Z < c_N) = 0.80$. Here x_T, x_N are contextual cutoffs and c_T, c_N are standardized cutoffs. For symmetric intervals the contextual endpoints are $-ac$ and ac under each model. Compare corresponding standardized endpoints separately. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

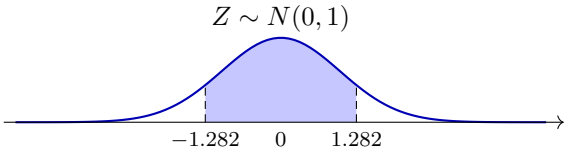
Mathematical reasoning and calculations

$$P(-c < T < c) = 0.8000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.1000.$$

Divide the outside complement $1 - p$ by two because symmetry makes the tails equal.

$$F_{29}(c) = 0.9000, \qquad c_T = F_{29}^{-1}(0.9) \approx 1.311, \quad c_N = \Phi^{-1}(0.9) \approx 1.282.$$





The standardized endpoints are $[-1.311, 1.311]$ under t and $[-1.282, 1.282]$ under normal. Compare the corresponding lower endpoints:

$$A = |-1.2815516 - (-1.3114336)| \approx 0.029882, \quad R = \frac{A}{1.3114336} \approx 0.022786, \quad E \approx 2.28\%.$$

Compare the corresponding upper endpoints:

$$A = |1.2815516 - (1.3114336)| \approx 0.029882, \quad R = \frac{A}{1.3114336} \approx 0.022786, \quad E \approx 2.28\%.$$

$$L_T = 0 - 20c_T \approx -26.23, \qquad U_T = 0 + 20c_T \approx 26.23 \quad \text{(units)}.$$

Normal gives $[-25.63, 25.63]$ units. Errors above compare standardized endpoints, not these contextual values. **Interpretation.** The t interval contains 80% of modeled deviations; 10% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.



3.5 Two-tail checks on filling weights

Problem. A quality team repeatedly collects groups of six filling weights. Individual measurements follow the bell-shaped pattern behind the first reference table, with its center and width adjusted to describe the measurements. The overall average is fixed and the overall spread is unknown. Measurements are chosen at random, and knowing one gives no information about another. For each group, find its average and take away the overall average. First find the positive number which, multiplied by itself, gives the number of readings. Divide the group's own measure of spread by this number, then divide the difference by that result. To obtain the group's measure of spread, multiply each reading's distance from the group average by itself, add these results, and divide by one less than the number of readings. The measure of spread is the positive number which, multiplied by itself, gives that last result. The group average and its measure of spread are calculated afresh for every group. Find two limits equally far below and above zero for the resulting readings, leaving ten percent of readings altogether beyond the two limits. Use the prediction rule appropriate to such groups. Compare it with the bell-shaped rule behind the first reference table, centered on zero with its original width, for the same requested share.

Questions/Tasks



C9 Describe the outcomes beyond the limits and the share at or below each limit. Find the limits using both rules. Compare the lower limits separately from the upper limits. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

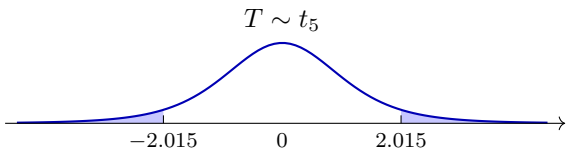
Let $n = 6$, μ be the population mean, σ the unknown population standard deviation, and $\bar{X}, S$ the sample mean and sample standard deviation. Independent random observations from a normal population imply $T = (\bar{X} - \mu)/(S/\sqrt{6}) \sim t_5$. Compare $Z \sim N(0,1)$. For positive cutoffs c_T, c_N, the target is $P(|T| > c_T) = P(|Z| > c_N) = 0.10$. Compare the standardized lower endpoints $-c_T, -c_N$ and upper endpoints c_T, c_N separately. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

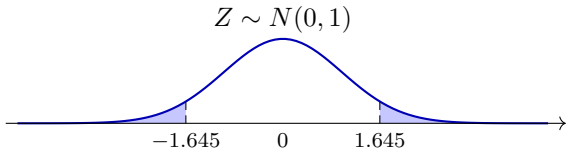
Mathematical reasoning and calculations

$$P(|T| > c) = 0.1000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0500.$$

Divide the combined outside area by two because symmetry makes the tails equal.

$$F_5(c) = 0.9500, \qquad c_T = F_5^{-1}(0.95) \approx 2.015, \quad c_N = \Phi^{-1}(0.95) \approx 1.645.$$





The standardized endpoints are $[-2.015, 2.015]$ under t and $[-1.645, 1.645]$ under normal. Compare the corresponding lower endpoints:

$$A = |-1.6448536 - (-2.0150484)| \approx 0.370195, \quad R = \frac{A}{2.0150484} \approx 0.183715, \quad E \approx 18.37\%.$$

Compare the corresponding upper endpoints:

$$A = |1.6448536 - (2.0150484)| \approx 0.370195, \quad R = \frac{A}{2.0150484} \approx 0.183715, \quad E \approx 18.37\%.$$

Interpretation. The t interval contains 90% of standardized sample-mean outcomes; 5% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.



3.6 Two-tail checks on processing times

Problem. A quality team repeatedly collects groups of twenty-five processing times. Individual measurements follow the bell-shaped pattern behind the first reference table, with its center and width adjusted to describe the measurements. The overall average is fixed and the overall spread is unknown. Measurements are chosen at random, and knowing one gives no information about another. For each group, find its average and take away the overall average. First find the positive number which, multiplied by itself, gives the number of readings. Divide the group's own measure of spread by this number, then divide the difference by that result. To obtain the group's measure of spread, multiply each reading's distance from the group average by itself, add these results, and divide by one less than the number of readings. The measure of spread is the positive number which, multiplied by itself, gives that last result. The group average and its measure of spread are calculated afresh for every group. Find two limits equally far below and above zero for the resulting readings, leaving four percent of readings altogether beyond the two limits. Use the prediction rule appropriate to such groups. Compare it with the bell-shaped rule behind the first reference table, centered on zero with its original width, for the same requested share.

Questions/Tasks



C9 Describe the outcomes beyond the limits and the share at or below each limit. Find the limits using both rules. Compare the lower limits separately from the upper limits. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting.

Worked solution

Translating the information into mathematics

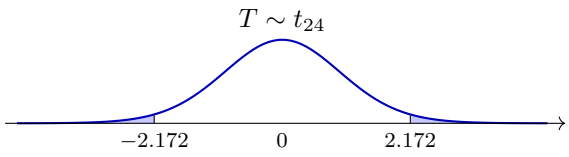
Let $n = 25$, μ be the population mean, σ the unknown population standard deviation, and $\bar{X}, S$ the sample mean and sample standard deviation. Independent random observations from a normal population imply $T = (\bar{X} - \mu)/(S/\sqrt{25}) \sim t_{24}$. Compare $Z \sim N(0,1)$. For positive cutoffs c_T, c_N, the target is $P(|T| > c_T) = P(|Z| > c_N) = 0.04$. Compare the standardized lower endpoints $-c_T, -c_N$ and upper endpoints c_T, c_N separately. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

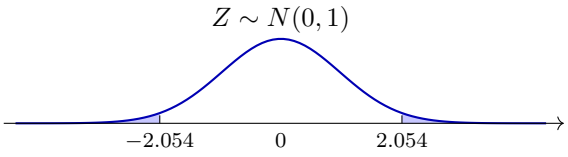
Mathematical reasoning and calculations

$$P(|T| > c) = 0.0400 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0200.$$

Divide the combined outside area by two because symmetry makes the tails equal.

$$F_{24}(c) = 0.9800, \qquad c_T = F_{24}^{-1}(0.98) \approx 2.172, \quad c_N = \Phi^{-1}(0.98) \approx 2.054.$$





The standardized endpoints are $[-2.172, 2.172]$ under t and $[-2.054, 2.054]$ under normal. Compare the corresponding lower endpoints:

$$A = |-2.0537489 - (-2.1715447)| \approx 0.117796, \quad R = \frac{A}{2.1715447} \approx 0.054245, \quad E \approx 5.42\%.$$

Compare the corresponding upper endpoints:

$$A = |2.0537489 - (2.1715447)| \approx 0.117796, \quad R = \frac{A}{2.1715447} \approx 0.054245, \quad E \approx 5.42\%.$$

Interpretation. The t interval contains 96% of standardized sample-mean outcomes; 2% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.



4 How sample size improves the approximation

Problem. A filling-weight audit repeatedly collects groups of six, twenty, sixty, and one hundred twenty weights. Individual measurements follow the bell-shaped pattern behind the first reference table, with its center and width adjusted to describe the measurements. The overall average is fixed and the overall spread is unknown. Measurements are chosen at random, and knowing one gives no information about another. For each group, find its average and take away the overall average. First find the positive number which, multiplied by itself, gives the number of readings. Divide the group's own measure of spread by this number, then divide the difference by that result. To obtain the group's measure of spread, multiply each reading's distance from the group average by itself, add these results, and divide by one less than the number of readings. The measure of spread is the positive number which, multiplied by itself, gives that last result. The group average and its measure of spread are calculated afresh for every group. For every group size, compare the prediction rule appropriate to such groups with the bell-shaped rule behind the first reference table, centered on zero with its original width.

Questions/Tasks

C9 For each group size and each rule, find the chance that the resulting reading is greater than two. Also find the upper limit exceeded by two and a half percent of readings. Compare the two chances and the two upper limits separately. For each comparison, report how far the second answer is from the first, how large that difference is compared with the size of the first answer, and that comparison as a percentage. Explain what the comparison means in this setting. Explain how the differences change as groups get larger and what this means for audit alerts.

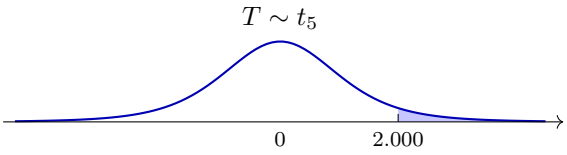
Worked solution

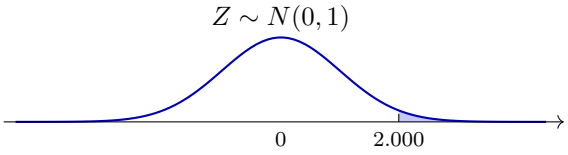
Translating the information into mathematics

Let $n \in \{6, 20, 60, 120\}$, μ denote the population mean, σ the unknown population standard deviation, and $\bar{X}, S$ the sample mean and sample standard deviation. Independent random observations from a normal population give $T = (\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$. Define $\nu = n - 1$ and $Z \sim N(0, 1)$. Compare $P(T > 2)$ with $P(Z > 2)$ and find c_T, c_N with $P(T > c_T) = P(Z > c_N) = 0.025$. Write the normal result as r_N and the t reference as r_T. Compute $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$ with unrounded values.

Mathematical reasoning and calculations

Keep the event fixed: $P(T > 2) = 1 - F_\nu(2)$ and $P(Z > 2) = 1 - \Phi(2)$. Keep the inverse target fixed: $P(T > c_T) = P(Z > c_N) = 0.025$, so use cumulative area 0.975. Here $\nu = n - 1$; the rows 59 and 119 require calculator supplements.





Probability comparison. Normal gives $1 - \Phi(2) \approx 0.0228$.

n	ν	t result	normal result	A	R	E (%)
6	5	0.0510	0.0228	0.028220	0.553654	55.37
20	19	0.0300	0.0228	0.007251	0.241688	24.17
60	59	0.0251	0.0228	0.002305	0.092000	9.20
120	119	0.0239	0.0228	0.001139	0.047665	4.77

Quantile comparison. Normal gives $\Phi^{-1}(0.975) \approx 1.960$.

n	ν	t result	normal result	A	R	E (%)
6	5	2.571	1.960	0.610618	0.237541	23.75
20	19	2.093	1.960	0.133060	0.063573	6.36
60	59	2.001	1.960	0.041031	0.020505	2.05
120	119	1.980	1.960	0.020136	0.010169	1.02

For example, in either table $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$, using unrounded calculator results. **Interpretation.** All three errors decrease across these sample sizes. Normal underestimates the chance of an audit statistic exceeding 2 and understates the cutoff required for a 2.5% alert rate. Heavier t tails explain both effects; increasing df reduces the difference. Estimating the population standard deviation still calls for t, even when the approximation is close.

